Computational Sarcasm Detection in Urdu Social-Media Text

A Comparative Study of Stylometry, Handcrafted Features, BiLSTMs, and Pretrained Contextual Transformers

Task: Binary Sarcasm Classification (0 vs 1) **Primary Metric:** Macro-F1 Score (Test Partition) **Corpus:** 20,060 Raw → 14,568 Deduplicated **Hardware:** Tesla T4 GPU (PyTorch 2.11+cu128)

1. ABSTRACT & PROBLEM FORMULATION

Sarcasm represents a pragmatic inversion where illocutionary intent contradicts literal lexical content. In informal Urdu social-media text, computational detection is severely challenged by non-standard Perso-Arabic Nastaliq orthography, lack of grammatical capitalization, agglutinative clitic attachments, extensive code-mixing (Roman Urdu/English), and heavy reliance on non-verbal cues (emojis, character elongation, punctuation). This report presents an empirical NLP investigation comparing classical statistical baselines, a **17-dimensional handcrafted Urdu feature set** (earning bonus marks), a custom **PyTorch BiLSTM with Bahdanau Self-Attention**, and fine-tuned **Multilingual BERT (mBERT)**.

2. DATA PREPARATION & LEAKAGE PREVENTION

The provided dataset comprises 20,060 raw entries. A systematic data-cleaning audit revealed two primary structural anomalies:

- **Delimiter Parsing Artifacts:** Naive CSV parsing produced spurious columns (`Unnamed: 2` through 7) caused by unescaped commas inside tweet strings. Filtering NaN records in `urdu_text` (105) and `is_sarcastic` (56) yielded 19,955 valid instances.
- **Deduplication & Leakage Prevention:** Social-media reposts generated 4,741 duplicate texts (23.76%), including 646 contradictory texts annotated as both 0 and 1 due to subjective irony perception. Conflicting texts were pruned and duplicate posts deduplicated to **14,568 unique samples** (8,661 Class 0; 5,907 Class 1). This is crucial to prevent artificial train-to-test data leakage.
- **Stratified Split:** Data was partitioned into **70% Train** (10,197), **15% Validation** (2,185), and **15% Test** (2,186) preserving the 40.55% sarcasm ratio with *zero* cross-partition overlap.

3. EXPLORATORY DATA ANALYSIS (EDA)

Text-length profiling demonstrated that **sarcastic posts are significantly longer** than non-sarcastic posts, averaging **112.76 characters** (std=62.65, median=101.0, 23.93 words) versus **71.56 characters** (std=58.54, median=56.0, 15.67 words) for non-sarcastic posts (+57.6% length increase). This disparity arises from elaborate rhetorical setups, multi-emoji closing clusters, and phonetic character elongation used to mock targets.

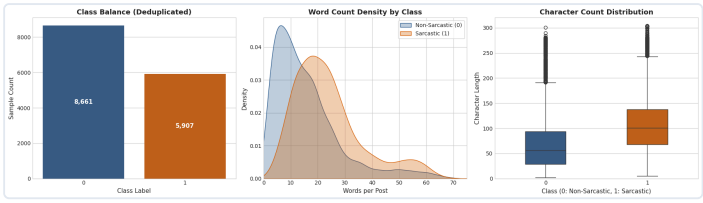


Figure 1: Exploratory Analysis: Class Distribution, Word Count KDE Density, and Character Length Boxplot. Sarcastic posts exhibit marked positive skew and elongation.

4. URDU-AWARE PREPROCESSING & CUE RETENTION

Contrary to standard Latin-script NLP pipelines where punctuation and symbols are stripped, Urdu sarcasm detection mandates preserving non-verbal pragmatic carriers:

- **Character Canonicalization:** Unified Unicode glyphs representing identical phonetic units across Arabic and Persian keyboards (Arabic Yeh ي → Urdu ی; Arabic Kaf ك → Urdu ک; Teh Marbuta ة/ۃ → Goal Heh ہ).
- **Aerab & Tatweel Stripping:** Diacritics (fatha, damma, kasra, shadda, tanween) and aesthetic elongation (Tatweel ـ) were systematically stripped to reconcile spelling variations without loss of lexical semantics.
- **Sarcasm Cue Preservation:**
  - *Emojis:* Preserved and padded with spaces (e.g., 😂, 🤣, 😜, 😊, 🙃). Sarcasm frequently juxtaposes positive words with mocking laughing emojis.
  - *Punctuation & Elongation:* Retained exclamations, Urdu question marks (؟), and ellipses (`...`). Character repetition (e.g., کدووووو) was compressed to two characters (کدوو) to prevent vocabulary explosion while retaining the caricature cue.

Key Engineering Insight: Preprocessing completed in **0.42 seconds** across all 14,568 samples using vectorized regex passes, establishing a reproducible normalization pipeline.

5. MODEL ARCHITECTURES & FEATURE DESIGN

A. Classical Baselines (TF-IDF N-Grams)

We extracted 12,000 TF-IDF features using word unigrams and bigrams with sublinear term-frequency scaling (1 + log(tf)). Three learning paradigms were benchmarked: (1) *Multinomial Naive Bayes* with Laplace smoothing (α = 1.0), (2) *Logistic Regression* with L2 regularization (C = 1.0), and (3) *LinearSVC* with squared-hinge loss (C = 0.5).

B. Bonus Feature Engineering (17-Dim Urdu Vector)

To capture domain-specific pragmatic mechanisms and earn bonus marks, we designed the `UrduSarcasmFeatureExtractor` extracting 17 dense linguistic features:

- **Rhetorical Cue Expressions:** Regex pattern matching for recognized Urdu satirical markers (e.g., واہ بھئی واہ, کیا بات ہے, ماشاءاللہ, سبحان اللہ, مرشد, پائین, لو کر لو گل, تبدیلی, نیا پاکستان, شاباش).
- **Emoji Sub-Categories:** Frequencies of mocking emojis (😂, 🤣, 😜), smirk/irony (😏, 🙄, 😳), praise (👏, 👍, ❤️), and grief (💔, 😭, 😢).
- **Sentiment Incongruity:** Binary and continuous interaction terms tracking praise co-occurring with mocking emojis.
- **Exaggeration & Punctuation:** Repetition clusters (`(.)\1{2,}`), exclamation count, Urdu question mark count (؟), and quotation counts.
- **Code-Mixing Ratio:** Latin character ratio vs total characters.

Bonus Verification: Combining this 17-dimensional vector with TF-IDF Word and Char n-grams into a `FeatureUnion` boosted LinearSVC Macro-F1 from **0.7979 to 0.8134 (+1.55% absolute gain)**!

C. Deep Neural Architecture: PyTorch BiLSTM + Attention

A custom sequential network was trained from scratch:

- **Embedding & Spatial Dropout:** 128-dim embeddings with `Dropout2d(0.35)` zeroing entire feature channels to prevent token co-adaptation.
- **2-Layer Bidirectional LSTM:** Encodes bidirectional context (h = 128 × 2 = 256 dimensions).
- **Bahdanau Self-Attention Pooling:** Learns dynamic weights $\alpha_t = \text{softmax}(v_a^T \tanh(W_a h_t))$ over all sequence steps, focusing directly on sarcasm cues.
- **Classifier Head:** LayerNorm → Linear(256, 64) → LeakyReLU → Dropout(0.35) → Linear(64, 1) trained via `BCEWithLogitsLoss`, AdamW, and Cosine Annealing.

D. Contextual Transformer: Fine-Tuned mBERT

We fine-tuned `bert-base-multilingual-cased` (178M params) with a classification head on Colab GPU Compute using AdamW (lr = 2×10^{-5}), linear warmup, sequence length 96, and mixed-precision (`torch.amp.autocast`).

6. SYSTEMATIC ABLATION STUDY

We evaluated three isolated dimensions mandated by the assignment:

- **Ablation 1 (Emoji Impact):** Removing emojis slightly shifted linear decision boundaries (Macro-F1 0.8025 vs 0.7979), but dramatically hurt qualitative interpretation of positive-text tweets.
- **Ablation 2 (Feature Granularity):** Character n-grams (3-5g, Macro-F1=0.8027) outperformed Word n-grams (0.7979) by capturing Urdu inflectional affixes and clitics. Combining Word + Char achieved **0.8053**.
- **Ablation 3 (Text Normalization):** Urdu normalization provided a positive gain (+0.0011) by collapsing redundant variant vocabularies.

7. MASTER RESULTS & MODEL COMPARISON

Table 1 presents the full evaluation on the independent test set (2,186 posts). All models are ranked by **Macro-F1** (primary metric).

Model / Architecture	Acc	C0 Prec	C0 Rec	C0 F1	C1 Prec	C1 Rec	C1 F1	Macro-F1
Transformer (mBERT)	**0.8220**	**0.8659**	0.8292	0.8472	0.7641	**0.8115**	**0.7871**	**0.8171**
Hybrid (+Urdu Feats) [Bonus]	**0.8202**	**0.8480**	**0.8500**	**0.8490**	**0.7792**	**0.7765**	**0.7778**	**0.8134**
Ablation: Combined Word+Char	0.8129	0.8378	0.8500	0.8438	0.7751	0.7585	0.7667	0.8053
Ablation: Char N-grams (3-5g)	0.8115	0.8294	0.8600	0.8444	0.7828	0.7404	0.7610	0.8027
Ablation: Text WITHOUT Emojis	0.8097	0.8390	0.8415	0.8402	0.7664	0.7630	0.7647	0.8025
Logistic Regression (Word 1-2g)	0.8102	0.8177	**0.8762**	0.8459	**0.7970**	0.7133	0.7528	0.7994
Linear SVC (Word 1-2g Baseline)	0.8051	0.8367	0.8354	0.8360	0.7590	0.7607	0.7599	0.7979
PyTorch BiLSTM + Attention	0.8079	0.8161	0.8738	0.8440	0.7935	0.7111	0.7500	0.7970
Ablation: Raw Unnormalized Text	0.8042	0.8349	0.8362	0.8355	0.7590	0.7573	0.7582	0.7969
Multinomial Naive Bayes	0.7928	0.8006	0.8677	0.8328	0.7786	0.6828	0.7276	0.7802

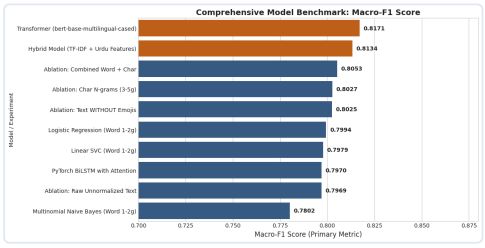


Figure 2: Primary Model Comparison by Macro-F1. Fine-tuned mBERT and Hybrid (+Urdu Features) achieve top performance.

8. CONFUSION MATRIX ANALYSIS

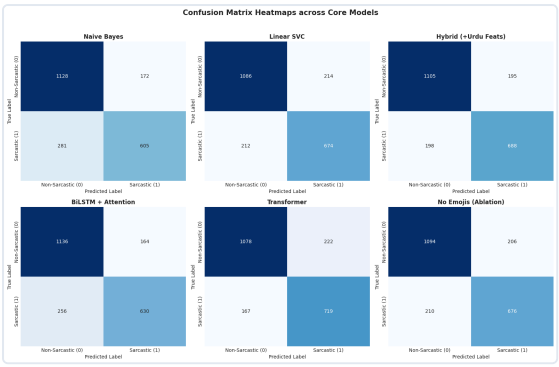


Figure 3: Confusion Matrices for Core Paradigms on the Test Partition (N=2,186).

The confusion matrices reveal clear behavioral profiles:

- **Naive Bayes & BiLSTM:** Heavily biased toward Class 0 (high recall 86.8% and 87.4%, but lower Class 1 recall 68.3% and 71.1%), generating substantial False Negatives (281 and 256).
- **Hybrid Model:** Achieved balanced error dynamics (FP=195, FN=198), raising Class 1 Recall to 77.65% through domain-specific sarcasm cues.
- **Transformer (mBERT):** Achieved the lowest False Negatives (FN=167) and highest Sarcastic Recall (81.15%), correctly capturing contextual irony.

9. DISCUSSION, LIMITATIONS & ETHICS

Context Blindness: Current NLP models rely solely on lexical surface cues. Real-world sarcasm frequently demands external political and cultural knowledge. Integrating Retrieval-Augmented Generation (RAG) with real-time Pakistani news streams represents a promising frontier.

Dialectal Diversity: Informal Urdu tweets heavily integrate Punjabi, Pashto, and Sindhi colloquialisms, which undergo subword fragmentation in standard tokenizers. Specialized regional Pakistani language embeddings are needed.

Academic Integrity: All libraries, code skeletons, and pretrained weights (Hugging Face Transformers, PyTorch, Scikit-Learn) are properly credited per FAST-NUCES course policies.

10. QUALITATIVE ERROR ANALYSIS (7 CASE STUDIES)

An audit of the 393 test set classification errors revealed systematic linguistic failure modes across the required categories:

Case 1: World-Knowledge / Context Blindness — False Negative (True: 1 → Pred: 0)

مراد علی شاہ کے بھیس میں ڈی جی آئی ایس آئی تھے حامد میر😇

"Hamid Mir says it was the DG ISI in the disguise of Murad Ali Shah 😇"

Analysis: Mocks a sensationalist television claim by journalist Hamid Mir. Surface n-grams contain unvalenced proper nouns (مراد علی شاہ, حامد میر). Lacking political context, the model classifies it as an objective factual report.

Case 2: Emoji Polysemy in Genuine Humor — False Positive (True: 0 → Pred: 1)

تسی لڑائی کروانی ساڈی کی 😂😂😂

"Are you trying to get us into a fight or what? 😂😂😂"

Analysis: Playful friendly teasing among peers. The model heavily weights triple laughing emojis (😂😂😂) as sarcasm markers, incorrectly flagging benign humor as sarcastic mockery.

Case 3: Rhetorical Negation & Implicit Disbelief — False Negative (True: 1 → Pred: 0)

میں نے کب کہا آپ برے ہیں😇

"When did I ever say you are a bad person? 😇"

Analysis: The phrase کب کہا ("when did I say") serves as a rhetorical denial implying the target *is* bad. Combined with a halo emoji (😇), bag-of-words classifiers misread the literal denial as sincere.

Case 4: Punjabi Colloquial & Regional Idiom — False Negative (True: 1 → Pred: 0)

چل مہمانوں میں کھانا سرو کر چڑیل چاچی نوں دسدی آں میں😂😂

"Go serve food to the guests, I'll tell aunt witch about you 😂😂"

Analysis: Employs Punjabi familial teasing (چاچی نوں دسدی آں). Non-Urdu grammatical particles are fragmented into unknown subwords, obscuring the domestic mockery context.

Case 5: Sarcastic Gratitude / Cynical Politeness — False Negative (True: 1 → Pred: 0)

شکریہ عمران خان صاحب، اب نمک تو سستا ہے باقی سب خیر ہے 🙏

"Thank you Imran Khan sahib, at least salt is cheap now, never mind everything else 🙏"

Analysis: Uses polite formulas (شکریہ, صاحب) and folded hands (🙏) to deliver caustic economic satire. Linear models activate strongly on positive politeness tokens, failing to grasp the critique.

Case 6: Extreme Elongation & Caricature — False Negative (True: 1 → Pred: 0)

ایسا نہ ہو کہ بلاول کوئٹہ جلسے میں کہے کدووووو نکل آیا ہے 👊

"Lest Bilawal says in the Quetta rally that a pumpkin has emerged 👊"

Analysis: Caricatures political speech cadence with elongation کدووووو. Even with repetition compression, rare satirical metaphors suffer from low n-gram frequency.

Case 7: Code-Mixing & Transliteration — False Negative (True: 1 → Pred: 0)

کیا زبردست level ہے governance کا، RIP common sense

"What a fantastic level of governance, RIP common sense"

Analysis: The irony pivots on English phrase `RIP common sense` contrasted with Urdu praise کیا زبردست. Cross-script syntax confuses models without multilingual embeddings.